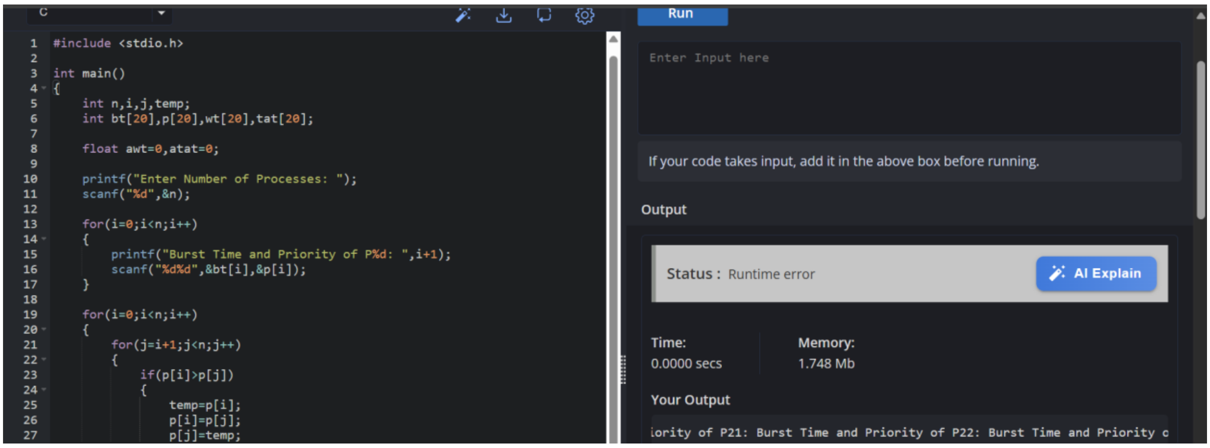
EXPERIMENT 05:

5. Construct a scheduling program with C that selects the waiting process with the highest priority to execute
next.



PSEUDO CODE :

Start

Input number of processes

Input Burst Time

Sort processes by Burst Time

Waiting Time of first process = 0

For remaining processes

Waiting Time = Sum of previous burst times

Turnaround Time = Waiting Time + Burst Time

Calculate averages

Display results

Start

Input number of processes

Input Burst Time

Input Priority

Sort by Priority

Waiting Time of first process = 0

Calculate Waiting Time

Calculate Turnaround Time

Calculate averages

Display results

Stop

SAMPLE OUTPUT :

Process BT Priority WT TAT

P2 3 1 0 3

P1 5 2 3 8

P3 4 3 8 12

Average Waiting Time = 3.67

Average Turnaround Time = 7.67

CODE:

```
#include <stdio.h>


int main()

{

  int n,i,j,temp;

  int bt[20],p[20],wt[20],tat[20];


  float awt=0,atat=0;


  printf("Enter Number of Processes: ");

  scanf("%d",&n);


  for(i=0;i<n;i++)

  {

    printf("Burst Time and Priority of P%d: ",i+1);

    scanf("%d%d",&bt[i],&p[i]);

  }


  for(i=0;i<n;i++)

  {

    for(j=i+1;j<n;j++)

    {

      if(p[i]>p[j])

      {

        temp=p[i];
```

```c
            p[i]=p[j];
            p[j]=temp;

            temp=bt[i];
            bt[i]=bt[j];
            bt[j]=temp;
         }
      }
   }

   wt[0]=0;

   for(i=1;i<n;i++)
      wt[i]=wt[i-1]+bt[i-1];

   for(i=0;i<n;i++)
   {
      tat[i]=wt[i]+bt[i];
      awt+=wt[i];
      atat+=tat[i];
   }

   printf("\nBT\tPriority\tWT\tTAT\n");

   for(i=0;i<n;i++)
      printf("%d\t%d\t\t%d\t%d\n",bt[i],p[i],wt[i],tat[i]);

   printf("\nAverage Waiting Time = %.2f",awt/n);
   printf("\nAverage Turnaround Time = %.2f",atat/n);

   return 0;
}
```